A PRINCIPAL PLANE SUBSCHEME IS A CURVE EXACTLY WHEN ITS EQUATION IS IRREDUCIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reducible construction, the weighted decomposition with component equations, the scalar comparison of generators, and the prime-divisor conclusion are all retained verbatim.

Editorial note. The label lemma-plane-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The plane-curve setup. Same chapter, Lemma lemma-equation-plane-curve. Fixes the closed subscheme Z of the projective plane with principal ideal generated by the homogeneous form F.
- Codimension-one divisors on the plane. Divisors, Lemmas divisors-lemma-codim-1-part and divisors-lemma-codimension-1-is-effective-Cartier. Provides the decomposition Z = sum a_i D_i into irreducible components with multiplicities, each D_i cut out by a homogeneous F_i.

Source reference. Original lemma 0AGB: sources/stacks-chow/tags.txt, source lines 13189–13189. Prerequisite pointer invoked at proof line(s) 1892; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BXH: sources/stacks-chow/tags.txt, source lines 14980–14980. Prerequisite pointer invoked at proof line(s) 1893; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0BYB: sources/stacks-curves/pinned-original-source.tex, source lines 1859–1859. Prerequisite pointer invoked at proof line(s) 1880; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Z \subset \mathbf{P}^2_k$ be as in Lemma 0BYB and let $I(Z) = (F)$ for some $F \in k[T_0, T_1, T_2]$. Then Z is a curve if and only if F is irreducible.*

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Proof. If F is reducible, say $F = F'F''$ then let Z' be the closed subscheme of $\mathbf{P}^2_k$ defined by F'. It is clear that $Z' \subset Z$ and that $Z' \neq Z$. Since Z' has dimension 1 as well, we conclude that either Z is not reduced, or that Z is not irreducible. Conversely, write $Z = \sum a_i D_i$ where D_i are the irreducible components of Z, see Divisors, Lemmas 0AGB and 0BXH. Let $F_i \in k[T_0, T_1, T_2]$ be the homogeneous polynomial generating the ideal of D_i. Then it is clear that F and $\prod F_i^{a_i}$ cut out the same closed subscheme of $\mathbf{P}^2_k$. Hence $F = \lambda \prod F_i^{a_i}$ for some $\lambda \in k^*$ because both generate the ideal of Z. Thus we see that if F is irreducible, then Z is a prime divisor, i.e., a curve. $\square$